Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.03	20.07	
worker nodes in service, mean	6.00	6.00	-0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.01	+0.2%
power (W), mean	627	643	+2.5%
power (W), peak	648	682	+5.3%
energy (Wh)	209	215	+2.6%
energy, parked workers still on at idle power (Wh)	209	215	+2.6%
CPU used (cores), mean	0.843	1.25	+48.4%
CPU allocatable (cores), mean	24.00	24.00	-0.0%
utilisation (used / allocatable)	0.035	0.052	+48.4%
energy per core-hour (Wh)	744	514	-31.0%
node-hours per core-hour	7.12	4.80	-32.6%
pending pods, pod-minutes	0.500	0.000	-100.0%
pending pods, peak	2.00	0.000	-100.0%
HPA replicas, mean	9.17	9.28	+1.2%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.517	0.533	+3.2%
requests timed	1,065	1,100	+3.3%
time over the response line (% of samples)	0.188	0.000	-100.0%
response time (ms), mean	127	95.12	-25.1%
response time (ms), median	114	82.20	-27.7%
response time (ms), 95th percentile	272	152	-44.1%
response time (ms), 99th percentile	358	193	-46.2%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.8563	5.2820	-32.8%	[-4.2993, -0.7402]	GAIN
kwh_per_core_hour	0.8174	0.5619	-31.3%	[-0.4264, -0.0805]	GAIN
utilisation	0.0347	0.0532	+53.5%	[+0.0051, +0.0328]	GAIN
pending_pod_minutes_per_hour	1.4400	0.0000	-100.0%	[-4.3200, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	1.4942	1.6931	+13.3%	[-2.7895, +2.7928]	STEADY, no significant difference

What Omni-Compass did

• decisions: 120 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	619	7.00	6.00	0.000	0.000	0.572	0.621
1	6.00	6.00	615	624	7.00	6.25	0.000	0.000	0.524	0.741
2	6.00	6.00	616	625	7.00	7.00	0.000	0.000	0.555	0.802
3	6.00	6.00	621	633	7.00	7.00	0.000	0.000	0.664	1.03
4	6.00	6.00	634	649	8.50	10.00	0.500	0.000	0.963	1.43
5	6.00	6.00	632	647	9.00	10.00	0.000	0.000	0.965	1.39
6	6.00	6.00	630	646	9.00	10.00	0.000	0.000	0.939	1.37
7	6.00	6.00	643	675	9.25	10.00	0.000	0.000	1.28	2.11
8	6.00	6.00	645	677	10.00	10.00	0.000	0.000	1.33	2.14
9	6.00	6.00	647	675	10.00	10.00	0.000	0.000	1.37	2.13
10	6.00	6.00	636	672	10.00	10.00	0.000	0.000	1.05	2.03
11	6.00	6.00	621	622	10.00	10.00	0.000	0.000	0.626	0.698
12	6.00	6.00	615	624	10.00	10.00	0.000	0.000	0.537	0.690
13	6.00	6.00	620	636	10.00	10.00	0.000	0.000	0.660	1.07
14	6.00	6.00	631	648	10.00	10.00	0.000	0.000	0.953	1.38
15	6.00	6.00	631	653	10.00	10.00	0.000	0.000	0.963	1.52
16	6.00	6.00	633	650	10.00	10.00	0.000	0.000	0.978	1.46
17	6.00	6.00	620	630	10.00	10.00	0.000	0.000	0.666	0.894
18	6.00	6.00	616	628	10.00	10.00	0.000	0.000	0.561	0.825
19	6.00	6.00	616	622	10.00	10.00	0.000	0.000	0.551	0.708

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.